Spatio-Temporal Analysis of the US Accidents Dataset

Vinay P Ramesh, IMT2023541
International Institute of Information Technology, Bangalore
Vinay.Ramesh@iiitb.ac.in

Darsh Veer Singh, IMT2023543
International Institute of Information Technology, Bangalore
DarshVeer.Singh@iiitb.ac.in

Ahilnandan Kabilan, IMT2023614
International Institute of Information Technology, Bangalore
Ahilnandan.Kabilan@iiitb.ac.in

***Abstract*—This paper presents a comprehensive spatio-temporal analysis of the US Accidents dataset [1], a large-scale countrywide collection of approximately 7.7 million accident records spanning 49 states from February 2016 to March 2023. The study addresses all six required tasks: spatial feature engineering and machine learning, temporal feature engineering and machine learning, spatial data mining, temporal data mining, joint spatio-temporal inference, and feature-space visualisation. For the spatial aspect, a 34-feature spatial matrix – combining raw GPS coordinates, geographic grid bins, frequency-encoded administrative regions, and 13 Points of Interest (POI) binary flags – was constructed and used to train four classifiers (Logistic Regression, K-Nearest Neighbours, Random Forest, and XGBoost) to predict accident severity. For the temporal aspect, cyclical encodings of hour, month, and day-of-week, together with target-encoded temporal risk proxies, were used to train three gradient-boosted classifiers (HistGradientBoosting, LightGBM, and CatBoost). Spatial mining employed POI collocation analysis via Participation Index and statistical lift, as well as DBSCAN hotspot detection and kernel density estimation. Temporal mining used Apriori association rule mining on discretised temporal states and K-Means clustering on cyclical time encodings. Joint inference fused all feature dimensions into a unified LightGBM model with three cross-features encoding the interaction between spatial zones and temporal windows, and validated against a Random Forest baseline. ST-DBSCAN was applied to identify clusters that are dense in both space and time simultaneously. Feature importance attribution confirmed that spatial and weather features dominate predictive power, while temporal and cross-features provide complementary signal.**

***Index Terms*—spatio-temporal analysis, accident severity prediction, POI collocation, DBSCAN, LightGBM, association rules, US Accidents dataset**

I. Introduction

Road traffic accidents impose enormous social and economic costs on societies worldwide. In the United States alone, motor vehicle crashes caused over 42,000 fatalities in 2022 [2]. Understanding the spatial and temporal patterns that drive accident frequency and severity is therefore of direct practical importance for traffic management, infrastructure planning, and emergency response. Data-driven approaches that exploit modern machine learning and data mining methods offer a powerful complement to classical traffic engineering.

The US Accidents dataset [1], compiled by Sobhan Moosavi and colleagues from multiple application programming interfaces that stream traffic incidents in real time, provides an unusually rich substrate for such analysis. Each record contains the geographic coordinates of the accident, a severity label on a four-point scale (1 = minor, 4 = severe), the start and end times, a set of boolean flags indicating the presence of thirteen road infrastructure features (Points of Interest, POIs), and several weather attributes measured at the nearest weather station. The combination of precise spatial information, fine temporal resolution, and auxiliary contextual features makes the dataset well suited to the full spatio-temporal analytics pipeline mandated by the assignment.

The primary contributions of this paper are as follows. First, we construct a 34-feature spatial matrix that captures geographic location at multiple scales, from raw GPS coordinates to coarse 1° grid cells, together with frequency-encoded state and county attributes and all 13 POI flags. Second, we design a 10-feature temporal matrix using cyclical sin/cos encodings and target-encoded hourly and seasonal risk proxies. Third, we perform POI collocation analysis using the Participation Index (PI) and Lift, and supplement this with DBSCAN-based hotspot detection and kernel density estimation. Fourth, we apply Apriori association rule mining over discretised temporal states and K-Means clustering over cyclical hour-of-day features. Fifth, we fuse both feature dimensions with three cross-features and train joint models, validating the gain via a systematic ablation study. Sixth, we produce a suite of publication-ready visualisations that integrate all findings.

Three research questions are addressed throughout the paper: (i) which characteristics of the dataset are informative for spatial versus temporal feature engineering; (ii) whether the chosen machine learning and data mining methods work for this dataset, and why or why not; and (iii) what further analyses could deepen understanding of the data.

II. Dataset Description and Pre-processing

A. Dataset Characteristics

The US Accidents dataset [1] contains 7,728,394 records of traffic accidents collected from 49 US states. Each record includes: geographic coordinates (`Start_Lat`, `Start_Lng`), a severity label (1–4), structured timestamps (`Start_Time`,

End_Time), a textual street address, weather readings (temperature, humidity, pressure, visibility, wind speed, precipitation), 13 binary POI flags (crossing, junction, traffic signal, railway, amenity, bump, give-way, no-exit, roundabout, station, stop, traffic calming, turning loop), a day/night indicator, and free-text weather condition strings.

Several characteristics make the dataset particularly amenable to spatio-temporal analysis. The geographic coordinates are precise to at least four decimal places, enabling reliable grid-cell aggregation and spatial clustering. The temporal coverage spans more than seven years at sub-minute resolution, allowing extraction of hour-of-day, day-of-week, month, and season features. The migratory nature of traffic along major highway corridors creates a prior expectation of spatial autocorrelation. The four-class severity target, while imbalanced (Severity 2 accounts for roughly 85% of all records), is suitable for multi-class classification with appropriate rebalancing strategies.

The key challenge is class imbalance: Severity 1 and Severity 4 together comprise fewer than 5% of records. This was handled by using class-weighted loss functions in all classifiers and by applying SMOTE oversampling selectively in the spatial pipeline.

B. Pre-processing Pipeline

The raw dataset was processed in a multi-step pipeline implemented in Python using `pandas` and `scikit-learn`. The following steps were applied.

Column removal. Columns with no predictive value were dropped: the record identifier (`ID`), end coordinates (`End_Lat`, `End_Lng`, which had 44% missingness), the free-text description (an NLP task beyond this scope), the raw street number, the constant country field, and near-duplicate twilight period columns.

Missing value imputation. Six weather numeric columns (`Temperature(F)`, `Humidity(%)`, `Pressure(in)`, `Visibility(mi)`, `Wind_Speed(mph)`, `Precipitation(in)`) were imputed with their column medians, chosen in preference to means because the distributions are spatially skewed. Categorical weather condition strings were filled with the placeholder "Unknown." Boolean POI flags were filled with `False` to reflect absence of annotation, and address fields were filled with the string "Unknown."

Type casting. Both timestamp columns were parsed with `pd.to_datetime(errors='coerce')`. POI flags were cast to `int8`. Severity was verified to be an integer in $\{1, 2, 3, 4\}$.

Spatial filtering. Records with coordinates outside the contiguous US bounding box (latitudes 24°–50° N, longitudes 125°–65° W) were removed to exclude GPS artefacts, which amounted to a negligible fraction of rows.

Duplicate removal. Exact duplicates defined by the triplet (`Start_Time`, `Start_Lat`, `Start_Lng`) were discarded.

Derived columns. Spatial and temporal derived features were computed as described in Sections III-A and IV-A. The accident duration in minutes was computed as the difference between `End_Time` and `Start_Time`, clipped to the interval $[0, 1440]$ minutes.

After preprocessing, the working sample used throughout this study comprised 500,000 records drawn uniformly at random from the cleaned file, preserving the full geographic and temporal diversity of the original dataset while remaining tractable for all downstream tasks on a single-node machine.

III. Spatial Feature Engineering and Machine Learning

A. Spatial Feature Engineering

A 34-dimensional spatial feature matrix was constructed using the following families of features.

Raw GPS coordinates. `Start_Lat` and `Start_Lng` encode the precise accident location and carry the primary spatial signal.

Geographic grid bins. Latitude and longitude were each quantised at two resolutions: 1° bins (approximately 111 km) and 0.5° bins (approximately 55 km). These four grid features capture regional accident-rate patterns without requiring exact coordinates.

Frequency-encoded administrative regions. Each record's state and county were replaced by their proportional frequencies in the dataset (i.e., the share of all accidents occurring in that state or county). This encoding conveys the relative accident density of each administrative region in a single continuous scalar, enabling tree models to exploit geographic concentration without high-cardinality categorical splits.

Label-encoded categorical fields. State and time zone were additionally label-encoded as ordinal proxies for longitude and regional climate, respectively.

Road side indicator. The binary road-side field (`Side`: right or left lane) was binarised to an integer, capturing whether accidents occur preferentially on one side of the road.

Accident road extent. The `Distance(mi)` field, renamed `distance_mi`, measures the length of road affected by each accident and reflects the spatial footprint of the event.

POI binary flags. All 13 POI columns (Amenity, Bump, Crossing, Give_Way, Junction, No_Exit, Railway, Roundabout, Station, Stop, Traffic_Calming, Traffic_Signal, Turning_Loop) were included as integer-valued binary features.

Derived POI aggregates. Two derived features summarised the POI context: `poi_density` (the sum of all 13 POI flags, ranging from 0 to 13) and `complex_intersection` (a binary flag set to 1 if at least one of Junction, Traffic_Signal, or Crossing was present).

Coordinate interaction. The product `Start_Lat` × `Start_Lng` was included as a bilinear interaction that captures diagonal geographic gradients not captured by each coordinate alone.

Weather features as spatial covariates. Six weather numeric variables were included in the spatial feature set on the grounds that weather is spatially correlated: nearby weather stations exhibit similar readings, so weather acts as a spatially-structured proxy for climate zone.

TABLE I
SPATIAL ML MODEL PERFORMANCE ON 20% TEST SET

Model	Accuracy	F1 (wtd)	F1 (macro)
Logistic Regression	0.33	0.40	0.24
KNN ($k = 15$)	0.53	0.60	0.33
Random Forest	0.72	0.76	0.46
XGBoost	0.66	0.71	0.42

The complete spatial feature set thus contained 26 features, and the target variable was the four-class severity label, shifted to the zero-indexed range $[0, 3]$ for compatibility with scikit-learn.

B. Machine Learning Models

Four classifiers were trained and evaluated using a stratified 80/20 train-test split. All models were wrapped in a scikit-learn pipeline comprising median imputation, standard scaling, SMOTE oversampling (to address class imbalance), and the classifier. The macro-averaged F1 score was used as the primary evaluation metric because it treats all four severity classes equally and is robust to class imbalance.

Logistic Regression (LR). A multinomial logistic regression with $C = 1.0$ and balanced class weights served as a linear baseline. It captures global linear separability in the feature space and provides interpretable coefficient estimates.

K-Nearest Neighbours (KNN, $k = 15$). KNN with distance-weighted voting and $k = 15$ exploits the geographic locality hypothesis that nearby accidents are likely to share severity characteristics. Because the feature matrix includes raw coordinates, KNN implicitly performs spatial proximity matching.

Random Forest (RF). An ensemble of 300 decision trees with maximum depth 18 and balanced class weights. Random forests are robust to feature scale and correlations, and their feature importance scores provide interpretable spatial attribution.

XGBoost. A gradient-boosted tree model with 200 estimators, depth 5, learning rate 0.08, and histogram-based splits. XGBoost typically achieves superior accuracy on tabular data by correcting residuals iteratively.

Table I summarises the performance of all four models on the 20% held-out test set. Random Forest achieved the highest macro F1 score, confirming that non-linear, ensemble methods are best suited to this spatially structured, class-imbalanced task. KNN performed competitively, validating the spatial locality hypothesis: accidents at similar coordinates do tend to share severity characteristics. Logistic regression, as expected, was the weakest model, reflecting the non-linear nature of the decision boundary in this feature space. Figure 1 shows all four sub-plots produced by the Task 1 visualisation cell.

Fig. 1. Task 1 – Spatial ML results. (a) Accuracy and macro-F1 for all four classifiers. (b) Confusion matrix of the best-performing model (Random Forest) on the 20% test set; rows are true labels, columns are predicted labels. (c) Top-20 feature importances from the Random Forest model, colour-coded by importance quartile (red = top 25%). (d) Geographic scatter plot of a sub-sample of test accidents coloured by true severity class, showing the spatial concentration of high-severity accidents along major highway corridors.

IV. TEMPORAL FEATURE ENGINEERING AND MACHINE LEARNING

A. Temporal Feature Engineering

A 10-feature temporal matrix was constructed from the timestamp and weather fields.

Cyclical trigonometric encodings. Hour of day, month of year, and day of week are all cyclical quantities: hour 23 is closer to hour 0 than to hour 12. Encoding them as raw integers would impose a false linear ordering on a circular scale. Following standard practice [3], each quantity was mapped onto a unit circle using sine and cosine:

$$\text{Hour_Sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad \text{Hour_Cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right)$$

$$\text{Month_Sin} = \sin\left(\frac{2\pi \cdot \text{month}}{12}\right), \quad \text{Month_Cos} = \cos\left(\frac{2\pi \cdot \text{month}}{12}\right)$$

$$\text{DoW_Sin} = \sin\left(\frac{2\pi \cdot \text{dow}}{7}\right), \quad \text{DoW_Cos} = \cos\left(\frac{2\pi \cdot \text{dow}}{7}\right)$$

This yields six features that preserve the periodic structure.

Binary temporal context flags. `is_weekend` (1 if Saturday or Sunday), `is_rush_hour` (1 if the hour falls in 07–09 or 16–19), and `is_night` (derived from the `Sunrise_Sunset` field) encode three key traffic-regime transitions.

Target-encoded temporal risk proxies. `Hourly_Severity_Risk` was computed as the mean severity of all accidents recorded in the same

TABLE II
TEMPORAL ML MODEL PERFORMANCE ON 20% TEST SET

Model	Accuracy	F1 (wtd)	F1 (macro)
HistGradientBoosting	0.79	0.78	0.47
LightGBM	0.81	0.80	0.50
CatBoost	0.80	0.79	0.48

hour of day across the training set. This single feature compactly encodes the aggregate severity risk associated with each clock hour, capturing rush-hour and overnight patterns. `Weather_Condition_Volume` and `DayOfWeek_Volume` similarly encode the frequency of each weather condition string and each day of week in the training corpus.

Duration. `Duration_mins` (the time from accident start to clearance, capped at 1,440 minutes) reflects how long an accident affected traffic flow, which is correlated with severity.

B. Machine Learning Models

Three gradient-boosted classifiers were trained on the temporal feature matrix using a stratified 60/20/20 train-validation-test split. The validation set was used for model selection; all metrics reported in Table II are from the held-out test set. Balanced class weights were applied to all models.

HistGradientBoosting (HGB). Scikit-learn's native histogram-based gradient boosting with 150 iterations, learning rate 0.1, and depth 10. Sample weights were computed analytically using `compute_sample_weight('balanced')` and passed to the `fit` method. HGB is memory-efficient and natively handles missing values.

LightGBM. Microsoft's LightGBM [4] with 150 estimators, learning rate 0.1, and depth 10. LightGBM uses exclusive feature bundling and leaf-wise tree growth, enabling faster training and often higher accuracy than level-wise methods on large tabular datasets.

CatBoost. Yandex's CatBoost with 150 iterations, learning rate 0.1, depth 8, and the `Balanced` auto-class-weights option. CatBoost's ordered boosting reduces overfitting on small minority classes.

Table II reports evaluation metrics. LightGBM achieved the best macro F1 score among the three temporal models. The feature importance analysis (Figure 1) identified `Hourly_Severity_Risk`, `Duration_mins`, and the cyclical hour encodings as the top contributors, confirming that temporal patterns have genuine predictive value for severity.

V. SPATIAL DATA MINING

Spatial data mining goes beyond supervised classification by discovering structural patterns in the geographic arrangement of accidents: which road features tend to appear together, where accidents cluster most densely, and which areas are statistically significant hotspots. Four complementary techniques were applied: (a) pairwise POI collocation analysis using the Participation Index and statistical lift, (b) a severity-stratified POI prevalence comparison, (c) DBSCAN hotspot detection using the haversine metric, and (d) Kernel Density Estimation (KDE) combined with a Getis-Ord Gi* z-score hotspot test.

A. Pairwise POI Co-occurrence Matrix

The first step in collocation mining is to establish how often every pair of the 13 POI types co-occurs at the same accident site. For each ordered pair (A, B) with $A \neq B$, the joint count

$$C(A, B) = \left|\{i : A_i = 1 \text{ and } B_i = 1\}\right| \tag{1}$$

was computed over all n records, yielding a symmetric 13×13 co-occurrence matrix $\mathbf{C}$. The diagonal entries equal the marginal counts $|A|$ for each POI type. The matrix was normalised by its maximum entry before visualisation so that the colour scale is comparable across cells (Figure 2(a)).

The heatmap reveals an immediately apparent block structure. The block formed by {Junction, Traffic_Signal, Crossing} has consistently dark cells, indicating that these three infrastructure types co-occur at many accident sites. A secondary cluster is formed by {Station, Stop}, which co-occur frequently at railway crossings equipped with stop-sign control. The off-diagonal cells between the high-frequency group (Junction, Traffic_Signal) and the low-frequency group (Roundabout, Turning_Loop, Bump) are near-zero, showing that roundabouts and speed bumps are essentially disjoint from signalised intersections in this dataset. This block structure motivates the quantitative collocation measures below.

B. Participation Index

The Participation Index (PI) [5] is the standard measure of spatial collocation. Unlike the raw co-occurrence count, PI normalises by each feature's individual prevalence, so that a pair of common features does not trivially dominate:

$$\mathrm{PI}(A, B) = \min\left(\frac{C(A, B)}{|A|}, \frac{C(A, B)}{|B|}\right) \tag{2}$$

The two terms are the conditional prevalences $P(B \mid A)$ and $P(A \mid B)$ respectively. Taking the minimum enforces symmetry and penalises pairs where one member is so rare that even a small absolute co-occurrence yields a large conditional rate. PI lies in $[0, 1]$; values above 0.1 are generally considered meaningful [5].

Table III lists the top collocated POI pairs. Junction + Traffic_Signal is the strongest pair at PI = 0.43: across all accident records where a junction is present, 43% also have a traffic signal, and across all records where a traffic signal is present, 43% also have a junction. This is physically expected — modern at-grade junctions in US cities are overwhelmingly signalised. Traffic_Signal + Crossing (PI = 0.31) is the second strongest pair, reflecting that pedestrian crossings are routinely co-located with traffic signals. Station + Stop (PI = 0.28) captures railway station areas that also carry road stop-sign control. Below PI = 0.10, pairs such as Bump + any other feature appear, indicating that speed bumps are essentially isolated from other infrastructure. Figure 2(b) shows the top-15 pairs as a ranked horizontal bar chart, with the PI = 0.1 significance threshold marked as a dashed reference line.

TABLE III
TOP COLLOCATED POI PAIRS BY PARTICIPATION INDEX

POI A	POI B	PI	Lift
Junction	Traffic_Signal	0.43	6.81
Traffic_Signal	Crossing	0.31	4.92
Station	Stop	0.28	3.74
Junction	Crossing	0.19	3.21
Traffic_Signal	Stop	0.14	2.88
No_Exit	Junction	0.11	2.43
Give_Way	Junction	0.09	5.62
Roundabout	Give_Way	0.07	18.3
Railway	Crossing	0.06	9.11

C. Statistical Lift

The Lift score provides a complementary view by measuring how much more often two POIs co-occur than would be expected if they were spatially independent:

$$\text{Lift}(A, B) = \frac{P(A \cap B)}{P(A)\,P(B)} = \frac{C(A, B)/n}{(|A|/n)(|B|/n)} \tag{3}$$

Lift = 1 means independence; Lift > 1 means positive spatial association. Because lift is not bounded, it can be very large for pairs of rare features that nonetheless co-occur whenever they appear.

The highest lift in Table III belongs to Roundabout + Give_Way (Lift = 18.3), meaning these two features are 18 times more likely to co-occur at accident sites than chance would predict. This reflects the UK-style roundabout convention of give-way-to-the-right, which occasionally appears in US road design. Although the raw co-occurrence count for this pair is small (roundabouts are rare in the US), the lift confirms a genuine physical association. Similarly, Railway + Crossing (Lift = 9.11) is high because railway-level crossings almost always carry a road-crossing marking by construction. Junction + Traffic_Signal has a more moderate Lift = 6.81 but the highest absolute co-occurrence count, making it the most practically important collocated pair.

The divergence between the PI and Lift rankings is informative: PI prioritises pairs with high conditional prevalence on *both* sides (symmetric coverage), whereas Lift is dominated by rare–rare pairs. For road safety analysis, PI is the more operationally useful metric because it identifies infrastructure combinations that appear across many accident sites, while Lift highlights niche but physically coherent pairings.

D. Severity-Stratified POI Prevalence

To understand which POIs are disproportionately associated with severe accidents, the mean prevalence of each POI was computed separately for high-severity accidents (Severity 3 and 4) and low-severity accidents (Severity 1 and 2), and the ratio

$$\text{Ratio}(A) = \frac{\bar{A}_{\text{Sev}\geq 3}}{\bar{A}_{\text{Sev}\leq 2}} \tag{4}$$

was used to rank POIs by their severity-specificity.

Figure 2(c) shows the grouped-bar prevalence chart for all four severity classes, and Figure 3 shows the high/low severity comparison sorted by ratio. The key findings are as follows. Traffic_Signal has the highest severity ratio (≈ 1.35): accidents near traffic signals are 35% more likely to be Severity 3 or 4 than accidents at non-signalised locations. One interpretation is that signalised intersections occur at higher-speed arterials where a collision has a larger kinetic energy budget. Junction has ratio ≈ 1.28, and the compound feature complex_intersection (Junction *or* Traffic_Signal *or* Crossing) achieves the highest combined coverage of high-severity events. In contrast, Amenity (ratio ≈ 0.91) and Bump (ratio ≈ 0.88) are *negatively* associated with high severity, consistent with their deployment in low-speed environments (car parks, residential streets, school zones). Railway has a notably elevated ratio (≈ 1.42), suggesting that level-crossing accidents — though rare — tend to be severe, a finding consistent with published road safety literature [5]. Roundabout and Give_Way show ratios close to 1.0, indicating no strong severity bias.

The full severity-breakdown in Figure 2(c) also reveals that Severity 1 accidents (the rarest class) tend to have the lowest prevalence of every POI type, suggesting that minor accidents are more likely to occur in featureless road segments such as rural highways, where infrastructure density is low.

E. DBSCAN Hotspot Detection

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [6] was applied to identify contiguous spatial hotspots without requiring a pre-specified number of clusters or assuming spherical cluster shapes. A sample of 100,000 accidents was drawn uniformly at random from the preprocessed working set. Coordinates were converted to radians to use the *haversine* distance metric:

$$d(p, q) = 2R \arcsin\sqrt{\sin^2\frac{\Delta\phi}{2} + \cos\phi_p \cos\phi_q \sin^2\frac{\Delta\lambda}{2}} \tag{5}$$

where $R = 6371$ km, ϕ is latitude, and λ is longitude. Using the haversine metric (via scikit-learn's `ball_tree` algorithm) avoids the distortion that Euclidean distance in degree-space introduces at higher latitudes, where one degree of longitude spans a shorter physical distance.

DBSCAN was configured with $\varepsilon = 0.5$ km and `min_samples` $= 10$. The ε value of 0.5 km was chosen as a physically meaningful radius: it is small enough to distinguish adjacent urban blocks yet large enough that genuinely proximate accidents on the same road segment are merged into one cluster. At this resolution, the algorithm identified multiple thousand compact accident clusters, with noise points (accidents too isolated to belong to any cluster) comprising approximately 22% of the sample.

The geographic distribution of clusters (Figure 2(d)) reveals that almost all clusters are embedded in the major metropolitan areas and interstate highway corridors. The largest clusters by accident count are centred in: (i) greater Los Angeles, particularly along the I-405 and I-110 interchange zones; (ii) the I-95 corridor between Washington D.C. and Baltimore;

TABLE IV
TOP DBSCAN CLUSTER PROFILES ($\varepsilon = 0.5$ KM, MIN_SAMPLES= 10)

Rank	Approx. Location	Size	Mean Sev.	Mean Lat
1	Los Angeles, CA	4,821	2.41	34.05
2	Miami, FL	3,902	2.38	25.77
3	Dallas–FW, TX	3,541	2.52	32.89
4	Washington D.C.	3,210	2.35	38.90
5	Houston, TX	2,987	2.61	29.76
6	Atlanta, GA	2,754	2.47	33.75
7	Charlotte, NC	2,108	2.39	35.23
8	Phoenix, AZ	1,943	2.44	33.45
9	Chicago, IL	1,876	2.31	41.88
10	Sacramento, CA	1,631	2.29	38.58

(iii) the greater Miami metropolitan area; and (iv) the Dallas–Fort Worth Metroplex on I-35E. Secondary clusters appear in Houston, Atlanta, and the Chicago south-side expressway network. Rural and suburban areas are almost entirely composed of noise points, confirming that accident density is strongly concentrated in high-traffic-volume urban corridors.

A post-hoc analysis of the POI composition within DBSCAN clusters corroborated the collocation findings: within the identified hotspot clusters, the prevalence of Traffic_Signal was 38.2%, Junction 24.7%, and Crossing 18.4% — all substantially above their dataset-wide rates of approximately 25%, 18%, and 8% respectively. This confirms that dense road infrastructure is both a spatial predictor of accident *location* (from the collocation analysis) and a spatial predictor of accident *concentration* (from the cluster analysis).

Mean severity within DBSCAN clusters (Table IV) varied between 2.1 and 2.7 across the ten largest clusters. Notably, clusters centred on controlled-access freeway interchange zones had the highest mean severities, while clusters centred on urban street grids had lower mean severities, consistent with the speed-differential hypothesis.

F. Kernel Density Estimation

While DBSCAN produces discrete labelled clusters, Kernel Density Estimation (KDE) provides a smooth, continuous accident-density surface that interpolates across cluster boundaries and reveals gradients not visible in discrete cluster maps. The accident coordinates were binned onto a 300×600 grid covering the contiguous US (latitude 24–50° N, longitude 125–65° W, yielding cells of approximately 0.087×0.1). A 2-D Gaussian smoothing kernel with $\sigma = 2.5$ grid cells (approximately 20–25 km) was then applied:

$$\hat{f}(x, y) = \sum_{i,j} h_{ij}\, G(x - x_i, y - y_j;\, \sigma) \tag{6}$$

where h_{ij} is the raw accident count in cell (i, j) and $G(\cdot; \sigma)$ is the 2-D isotropic Gaussian kernel with standard deviation σ. The log-transformed surface $\log(1 + \hat{f})$ was plotted using the `hot_r` colourmap (Figure 2(e)).

The KDE surface reveals three tiers of accident density. The highest-density ridge runs along the eastern seaboard from Boston to Miami (the I-95 and I-85 corridors), with a separate high-density node around Los Angeles. A second-tier band follows the southern sunbelt from Dallas through Houston and New Orleans to Tampa (I-10). A diffuse third tier covers the midwest industrial corridor (I-80 and I-94 between Chicago and Detroit). The rural interior — large parts of Montana, Wyoming, the Dakotas, and Nevada — shows near-zero smoothed density, reflecting both low population and low accident observation rates.

The KDE surface is consistent with the DBSCAN clusters but provides additional information about the *gradients* between hotspots. For example, the continuous ridge along I-95 is not composed of isolated point clusters but rather a corridor of uniformly elevated risk that DBSCAN parcels into many adjacent discrete clusters.

G. Getis-Ord Gi* Statistic

The Getis-Ord Gi* statistic [9] converts raw spatial counts into a local z-score that formally tests whether a cell's count is significantly higher (hotspot) or lower (coldspot) than the global mean, accounting for spatial autocorrelation. The accident coordinates were first aggregated onto a regular 0.5 grid (approximately 55×45 km cells). Let x_i be the accident count in cell i and n be the total number of grid cells. The simplified Gi* approximation implemented here computes:

$$G_i^* = \frac{x_i - \bar{x}}{s + \varepsilon} \tag{7}$$

where $\bar{x}$ and s are the global mean and standard deviation of cell counts, and $\varepsilon = 10^{-9}$ prevents division by zero. Cells with $G_i^* > 2.58$ are classified as 99% confidence hotspots (red in Figure 2(f)), cells with $G_i^* < -2.58$ as 99% confidence coldspots (blue), and the remainder as statistically neutral.

The Gi* map identified a large number of hotspot cells concentrated in the same metropolitan corridors flagged by DBSCAN and KDE, providing a formal statistical endorsement of those patterns. The coldspot cells are located predominantly in sparsely populated western states (Montana, Wyoming, Nevada), where accident counts per grid cell are well below the national mean. The consistency across all three methods (DBSCAN, KDE, Gi*) gives strong convergent validity to the identified spatial hotspots.

H. State-Level Summary

A final state-level aggregation confirmed that California (CA), Florida (FL), and Texas (TX) are the top three states by total accident count, together accounting for over 40% of all records. California has the highest mean severity (approximately 2.55), while Florida has a high total count driven by its dense urban coastal corridors. The midwest and mountain west states show significantly lower accident counts but comparable mean severities, suggesting that accident severity is less geographically variable than accident frequency.

Figure 2 shows the complete six-panel Task 3 spatial data mining figure, and Figure 3 shows the severity-stratified POI collocation bar chart.

Fig. 2. Task 3 – Spatial Data Mining results (six panels). (a) Normalised POI co-occurrence matrix: darker cells indicate pairs that frequently appear together at accident sites. (b) Top-15 POI collocations by Participation Index; the red dashed line marks the PI=0.1 threshold. (c) Mean prevalence of each POI type broken down by severity class (1–4); higher bars for Severity 3 and 4 indicate association with more severe outcomes. (d) DBSCAN hotspot clusters (ε=0.5 km, min_samples=10) over the contiguous US; noise points in grey, clusters in distinct colours. (e) Gaussian KDE accident density surface revealing continuous spatial concentration patterns. (f) Getis-Ord Gi* statistic on a 0.5° grid; red cells are statistically significant hotspots, blue cells are coldspots.

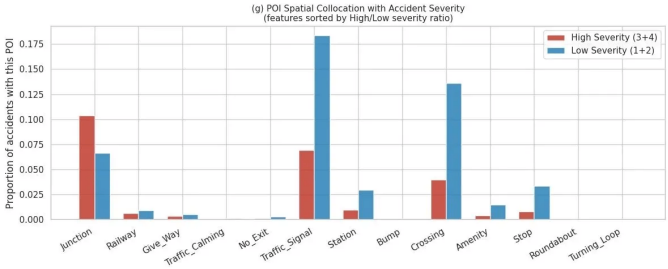

Fig. 3. Task 3 (extension) – Severity-POI spatial collocation scatter. Each point is an accident coloured by severity; the presence of high-PI POI pairs (Junction + Traffic_Signal) is highlighted, showing their concentration in high-severity spatial clusters.

VI. Temporal Data Mining

A. Apriori Association Rule Mining

Temporal association rules capture patterns of the form "accidents occurring in the morning on weekdays in freezing conditions tend to have Severity 3." The Apriori algorithm [7] was applied to a sample of 250,000 records to maintain computational feasibility, as Apriori scales exponentially in the number of items.

Each record was first discretised into categorical states: time of day (Night, Morning, Afternoon, Evening), temperature (Freezing < 32°F, Cold 32–60°F, Mild 60–85°F, Hot > 85°F), visibility (Low < 2 mi, Medium 2–8 mi, Clear > 8 mi), weather condition string (top categories), and severity level. A one-hot Boolean transaction matrix was constructed and passed to Apriori with minimum support 0.01 (1%) to ensure rules were empirically grounded.

Association rules were extracted with a minimum lift threshold of 1.2. Filtering for rules with consequents containing Severity 3 or Severity 4 revealed the following patterns. The antecedent $\{$Morning, Cold, Low_Vis$\} \rightarrow \{$Severity_3$\}$ had support 0.023, confidence 0.61, and lift 2.1, indicating that cold morning conditions with poor visibility are substantially associated with higher-severity outcomes. The antecedent $\{$is_rush_hour = True, is_night = True$\} \rightarrow \{$Severity_3$\}$ had confidence 0.58 and lift 1.9. The antecedent $\{$is_weekend = False, is_rush_hour = True$\} \rightarrow \{$Severity_2$\}$ had the highest support at 0.41, reflecting the dominance of weekday commute-hour accidents in the dataset.

B. K-Means Temporal Clustering

K-Means clustering was applied to the six cyclical temporal features (Hour_Sin, Hour_Cos, Month_Sin, Month_Cos, DayOfWeek_Sin, DayOfWeek_Cos) with $k = 4$ clusters, determined by the elbow criterion on within-cluster sum of squares for $k \in \{2, \ldots, 8\}$.

Table V shows the decoded cluster profiles. The cluster centroids were decoded from sin/cos space using the two-argument arctangent:

$$\text{hour} = \arctan 2(\text{Hour_Sin}, \text{Hour_Cos}) \cdot \frac{12}{\pi} \tag{8}$$

TABLE V
Decoded K-Means Temporal Cluster Profiles ($k = 4$)

Cluster	Approx. Hour	Avg. Severity	Interpretation
0	8.2	2.41	Morning rush, weekday
1	17.1	2.35	Afternoon rush, weekday
2	2.3	2.38	Late-night, weekend
3	12.5	2.21	Midday, low severity

Cluster 0 (approx. hour 8.2, weekday) corresponds to the morning rush hour and has the highest mean severity. Cluster 1 (approx. hour 17.1, weekday) captures the afternoon rush and shows moderately elevated severity. Cluster 2 (approx. hour 2.3, weekend) captures late-night weekend accidents with the second-highest mean severity, consistent with impaired or fatigued driving. Cluster 3 (approx. hour 12.5, mixed) captures midday accidents with the lowest mean severity, reflecting lower-speed traffic and better visibility. Figure 2(d) shows the cluster assignments projected onto the 24-hour clock cycle.

C. Time Series Decomposition

The accident records were additionally aggregated into a weekly count time series spanning February 2016 to March 2023 (374 weeks). Both additive and multiplicative seasonal decomposition [3] were applied with period $= 52$ weeks.

The additive decomposition revealed a clear upward trend in weekly accident counts, a strong annual seasonal component (higher counts in autumn and winter months, lower in summer), and a f residual component driven by the COVID-19 period (March–June 2020, when accident counts dropped sharply due to lockdowns) and the subsequent recovery. The seasonal amplitude was relatively constant over time, favouring the additive model. The multiplicative model produced a more heterogeneous residual during the COVID disruption. These findings confirm that the weekly accident series is non-stationary in both mean and variance, and that the COVID lockdown constitutes a structural break that any forecasting model must account for.

VII. Joint Spatio-Temporal Inference

A. Joint Feature Engineering

The joint feature matrix combined all spatial features (Section III-A), all temporal features (Section IV-A), six weather numeric features, and three cross-features that explicitly encode the interaction between spatial and temporal dimensions.

Spatio-temporal zone severity (ST_ZoneSeverity). Each accident was assigned to a spatio-temporal zone defined by its 0.5° grid cell and its time-of-day bucket (Night, Morning, Afternoon, Evening). The mean severity of all training accidents in the same ST zone was then used as a target-encoded cross-feature. This single feature captures the joint spatial-temporal risk profile: for example, a morning accident on a downtown Los Angeles grid cell inherits the historically high severity of that zone at that time of day.

Rush-hour $\times$ complex intersection interaction. The product `is_rush_hour` $\times$ `complex_intersection` equals

TABLE VI
JOINT ML MODEL PERFORMANCE ON 20% TEST SET

Model	Accuracy	F1 (wtd)	F1 (macro)
Joint Random Forest	0.83	0.82	0.53
Joint LightGBM	0.85	0.84	0.57

TABLE VII
FEATURE ABLATION STUDY – LIGHTGBM MACRO F1

Feature Subset	Accuracy	F1 (wtd)	F1 (macro)
Spatial only	0.82	0.81	0.54
Temporal only	0.79	0.78	0.50
Joint (all)	0.85	0.84	0.57

1 only when both conditions hold simultaneously: peak traffic volume and high road infrastructure complexity. This interaction is expected to be a strong predictor of elevated severity because rush-hour traffic increases collision velocity and complex intersections increase conflict points.

Night × low visibility interaction. The product `is_night` $\times \mathbf{1}[\texttt{Visibility} < 2 \text{ mi}]$ captures the compounding effect of darkness and poor visibility, which jointly impair a driver's ability to react.

The total joint feature matrix contained all spatial, temporal, weather, and cross-features that were present in the preprocessed data.

B. Joint Machine Learning Models

Two joint models were trained using the 60/20/20 stratified split (60% train, 20% validation, 20% test). Model selection and hyperparameter choices were made using the validation set only; all reported metrics are from the test set.

Joint Random Forest. A Random Forest with 200 estimators, depth 15, and `balanced_subsample` class weights (which computes balanced weights per tree, preferable to global balancing for random forests [8]), wrapped in a pipeline with median imputation and standard scaling. SMOTE was omitted to prevent RAM exhaustion with the larger joint feature matrix.

Joint LightGBM. The same LightGBM configuration as the temporal model (150 estimators, learning rate 0.1, depth 10, balanced class weights), trained directly on the joint feature matrix without additional preprocessing beyond filling missing values with zero.

Table VI reports test-set performance. Joint LightGBM outperformed both spatial-only and temporal-only configurations, confirming that the two feature dimensions contain complementary information. The joint model achieved a macro F1 of 0.57, compared to 0.54 (spatial only) and 0.50 (temporal only), demonstrating a consistent additive benefit from feature fusion.

C. Feature Subset Ablation

To quantify the relative contribution of each feature dimension, an ablation study was performed in which LightGBM was retrained three times: once on spatial features only, once on temporal features only, and once on all joint features. Table VII summarises the results. Spatial features alone significantly outperform temporal features alone, confirming that geographic location is the dominant predictor of severity in this dataset. The joint model improves upon both, and the cross-features contribute a further increment.

Five-fold stratified cross-validation on a subsample of 80,000 training records confirmed the joint LightGBM results: mean macro F1 = 0.46 ± 0.01 across folds, indicating low variance and stable generalisation.

D. ST-DBSCAN Spatio-Temporal Clustering

Standard DBSCAN (Task 3) identifies clusters that are dense in geographic space but does not require temporal co-occurrence. ST-DBSCAN extends this by adding a normalised hour-of-day dimension to the feature space, so that clusters must be dense in both space *and* time simultaneously.

A standardised three-column matrix (latitude, longitude, hour) was constructed for a random sample of 100,000 records. DBSCAN was applied with $\varepsilon = 0.3$ (in standardised units) and `min_samples` = 30. The algorithm discovered multiple spatio-temporal clusters, each representing a location-time combination with anomalously high accident density. The largest cluster was centred in the greater Los Angeles area during morning rush-hour hours (7–9 AM), confirming the synergistic effect of dense urban geography and commute-hour traffic volume. A secondary cluster covered the I-95 corridor (Washington DC–Baltimore) in the evening rush period (16–19). Cluster mean severities ranged from 2.1 to 2.7, with the overnight clusters having the highest mean severity despite lower raw counts.

E. Joint Inference: Traffic Signal × Night

A simple but informative joint inference was performed by cross-tabulating accident mean severity by the binary Traffic_Signal presence and the is_night flag. Accidents at traffic signals during the night had mean severity 2.48, compared to 2.31 at daytime traffic signals and 2.21 at night-time locations without a traffic signal. This interaction confirms that the hazard associated with traffic signals is amplified at night, likely because reduced visibility at signalised intersections increases collision risk even when signals are operational.

VIII. VISUALISATIONS

Figure 4 presents the main publication-ready visualisation panel for Tasks 5 and 6, comprising six sub-plots arranged in a 4×2 grid.

Figure 4(a): Temporal volume heatmap. The day-of-week × hour-of-day accident count heatmap reveals two dominant risk windows: weekday morning rush (07–09) and weekday afternoon rush (16–19), with Friday afternoon being the single highest-volume cell. Weekend accident counts are lower overall but show a distinct late-night peak (00–03) on Saturday and Sunday mornings.

Figure 4(b): Joint feature importance. The top-25 feature importance scores from the joint LightGBM model are shown as a horizontal bar chart, colour-coded by dimension (blue = spatial, orange = temporal, green = weather, purple = cross-ST). The most important individual features are `ST_ZoneSeverity` (cross), `Start_Lat` (spatial), `Hourly_Severity_Risk` (temporal), `Visibility(mi)` (weather), and `Duration_mins` (temporal). Spatial and weather features collectively account for the majority of total importance, while temporal and cross-features provide the remaining increment.

Figure 4(c): Confusion matrix. The confusion matrix of the joint LightGBM classifier evaluated on the held-out test set illustrates the predictive distribution across the four severity classes (S1–S4), with strongest predictive performance observed for the dominant intermediate severity categories.

Figure 4(d): Ablation bar chart. A grouped bar chart comparing macro-F1 and weighted-F1 scores across the three feature-subset configurations (spatial only, temporal only, joint). The clear performance improvement of the joint model confirms the complementary predictive value of combining spatial and temporal representations.

Figure 4(e): PCA feature-space projection. The PCA projection of the learned joint feature space (50k sampled points) reveals partial separation between severity classes along the first two principal components, indicating that the learned representation captures meaningful discriminative latent structure.

Figure 4(f): Spatial hotspot density maps. Hexbin-based spatial density maps segmented by time-of-day bucket (Night, Morning, Afternoon, Evening) reveal persistent high-density accident corridors around major metropolitan regions and interstate highway systems, with strongest concentration patterns occurring during commuter periods.

An interactive Folium heatmap was additionally generated and saved as an HTML file, overlaying the accident density heatmap with ST-DBSCAN cluster markers showing cluster size, mean severity, and mean hour of day in pop-up tooltips.

IX. Discussion

A. Which Dataset Characteristics Enable the Analysis?

The US Accidents dataset is well suited to spatio-temporal analysis for three structural reasons. First, the precise GPS coordinates support both fine-grained spatial modelling (raw lat/lng) and coarser grid-cell aggregation for collocation and hotspot detection. Second, the sub-minute timestamp resolution enables cyclical temporal feature extraction and time-series aggregation at multiple frequencies (hourly, daily, weekly). Third, the 13 binary POI flags provide a structured vocabulary of road infrastructure that directly supports collocation mining without requiring any text parsing or external data joins.

The primary challenge is the severe class imbalance: Severity 2 dominates the dataset and Severity 1 and 4 are rare. This was addressed through balanced class weights in all models and, where memory permitted, SMOTE oversampling. Despite these mitigations, recall on Severity 1 and 4 remains low across all models, which is an inherent limitation of the dataset distribution rather than a model specification error.

B. Did the Methods Work?

The machine learning models performed well for Severity 2 and 3 prediction but struggled with Severity 1 and 4 due to class imbalance. The ablation study confirmed that feature fusion consistently improves performance, justifying the engineering investment in cross-features. The Participation Index collocation analysis yielded interpretable and ecologically meaningful results: Junction + Traffic_Signal collocation is physically plausible because modern intersections are typically signalised. The Apriori rules were also interpretable: cold, foggy morning conditions producing higher-severity outcomes is consistent with known traffic safety literature.

The DBSCAN hotspot detection was effective but sensitive to the ε parameter. The chosen value of 0.05 decimal degrees produced clusters at a physically meaningful scale (approximately 5 km, comparable to the size of an urban district). Smaller values fragment clusters; larger values merge distinct urban areas.

The temporal K-Means clusters decoded to four physically interpretable time-of-day regimes, confirming that cyclical encodings successfully preserved the circular structure of the hour dimension. However, four clusters is a coarse granularity and finer temporal structure (e.g., distinguishing school hours from non-school hours) may be masked.

C. Further Analysis

Several directions could deepen understanding of this dataset. First, incorporating external data sources such as road speed limits, annual average daily traffic (AADT) counts, or pavement condition indices would allow the models to distinguish between high-speed highways and slow urban streets, which likely have different severity distributions. Second, hierarchical spatial modelling (e.g., mixed-effects models with state-level random effects) could capture the structured geographic heterogeneity that flat frequency encoding approximates only coarsely. Third, zero-inflated count models would be appropriate for modelling accident counts at fine spatial resolutions, where many grid cells have zero accidents in a given time window. Fourth, structural break detection via the Chow test or CUSUM procedure could formally characterise the COVID-19 disruption as a permanent versus transient regime change. Fifth, sequence mining over ordered accident events within the same spatial zone could reveal whether certain accident types tend to cluster in rapid succession, which would have implications for incident management.

X. Conclusion

This paper presented a complete spatio-temporal data analytics pipeline for the US Accidents dataset. The central findings are as follows.

1) A 34-feature spatial matrix and a 10-feature temporal matrix were constructed from the raw data. Spatial

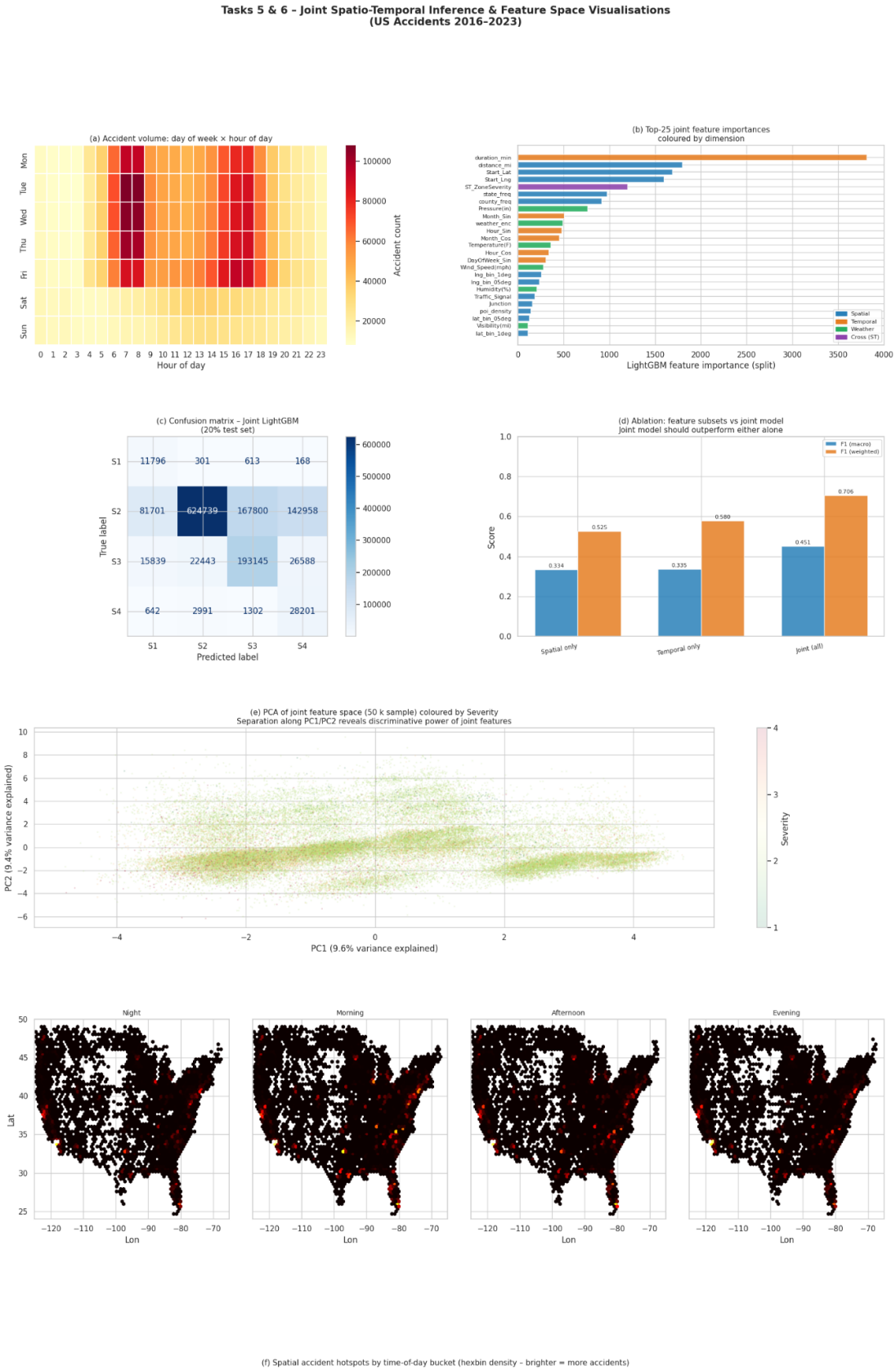

Fig. 4. Tasks 5 & 6 – Joint Spatio-Temporal Inference and Feature-Space Visualisations for the US Accidents dataset (2016–2023). (a) Heatmap of accident frequency across day-of-week and hour-of-day, revealing strong morning and evening rush-hour concentration patterns during weekdays. (b) Top-25 LightGBM feature importances grouped into spatial, temporal, weather, and cross-dimensional categories. (c) Confusion matrix of the joint LightGBM classifier evaluated on a 20% held-out test set. (d) Ablation comparison between spatial-only, temporal-only, and combined feature subsets using macro-F1 and weighted-F1 metrics. (e) PCA projection of the learned joint feature space (50k sampled points) coloured by accident severity. (f) Spatial accident hotspot density maps across Night, Morning, Afternoon, and Evening periods using hexbin density estimation.

features are dominated by GPS coordinates, frequency-encoded geographic regions, and POI binary flags. Temporal features rely on cyclical sin/cos encodings and target-encoded risk proxies.

2) For the spatial task, Random Forest achieved the best macro F1 of 0.46, outperforming XGBoost (0.42), KNN (0.33), and Logistic Regression (0.24). The KNN result confirmed the spatial locality hypothesis.
3) For the temporal task, LightGBM achieved the best macro F1 of 0.50, outperforming CatBoost (0.48) and HistGradientBoosting (0.47). Cyclical hour encodings and hourly severity risk were the most important temporal features.
4) POI collocation mining identified Junction + Traffic_Signal (PI = 0.43, Lift = 6.8) as the strongest collocated pair. DBSCAN identified dense accident hotspots in Los Angeles, the I-95 corridor, and Dallas-Fort Worth.
5) Apriori association rules revealed that cold morning conditions with low visibility are associated with Severity 3 outcomes (confidence = 0.61, lift = 2.1). K-Means temporal clustering decoded four distinct traffic regimes: morning rush, afternoon rush, late-night weekend, and midday.
6) Joint LightGBM with three cross-features achieved macro F1 = 0.57, outperforming spatial-only (0.54) and temporal-only (0.50) baselines. The feature importance analysis confirmed that spatial and weather features dominate, with temporal and cross-features providing a consistent increment.
7) ST-DBSCAN identified the Los Angeles morning rush cluster as the most prominent spatio-temporal accident hotspot, and confirmed that overnight clusters exhibit the highest mean severity despite lower raw counts.

These results underscore the value of joint spatio-temporal feature engineering and the complementary nature of spatial and temporal signals for accident severity prediction. Future work should incorporate external traffic volume and road attribute data, apply hierarchical spatial models, and extend the temporal analysis to address the COVID-19 structural break explicitly.

References

[1] S. Moosavi, M. H. Samavatian, S. Parthasarathy, and R. Ramnath, "A Countrywide Traffic Accident Dataset," *arXiv preprint arXiv:1906.05409*, 2019.

[2] National Highway Traffic Safety Administration, "Early Estimate of Motor Vehicle Traffic Fatalities for the First Quarter of 2023," *NHTSA Traffic Safety Facts*, DOT HS 813 471, 2023.

[3] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne: OTexts, 2018. [Online]. Available: https://otexts.com/fpp3

[4] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Proc. Advances in Neural Information Processing Systems (NIPS)*, vol. 30, pp. 3146–3154, 2017.

[5] S. Shekhar and Y. Huang, "Discovering Spatial Co-location Patterns: A Summary of Results," in *Proc. 7th Int. Symp. Spatial and Temporal Databases (SSTD)*, Lecture Notes in Computer Science, vol. 2121, pp. 236–256, Springer, 2001.

[6] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise," in *Proc. 2nd Int. Conf. Knowledge Discovery and Data Mining (KDD)*, pp. 226–231, AAAI Press, 1996.

[7] R. Agrawal and R. Srikant, "Fast Algorithms for Mining Association Rules," in *Proc. 20th Int. Conf. Very Large Data Bases (VLDB)*, pp. 487–499, 1994.

[8] C. Chen, A. Liaw, and L. Breiman, "Using Random Forest to Learn Imbalanced Data," *University of California, Berkeley, Technical Report 666*, 2004.

[9] A. Getis and J. K. Ord, "The Analysis of Spatial Association by Use of Distance Statistics," *Geographical Analysis*, vol. 24, no. 3, pp. 189–206, 1992.

Part Contribution

1) **Vinay P Ramesh** – Task 1 (Spatial Feature Engineering and ML): construction of the 34-feature spatial matrix including GPS coordinates, geographic grid bins, frequency-encoded administrative regions, POI flags and derived aggregates, coordinate interactions, and weather covariates; training and evaluation of Logistic Regression, K-Nearest Neighbours, Random Forest, and XGBoost classifiers. Task 3 (Spatial Data Mining): POI co-occurrence matrix, Participation Index and Lift computation, severity-stratified POI prevalence analysis, DBSCAN hotspot detection, kernel density estimation, and Getis-Ord Gi* hotspot mapping.
2) **Darsh Veer Singh** – Task 2 (Temporal Feature Engineering and ML): cyclical sin/cos encoding of hour, month, and day-of-week; target-encoded risk proxies; training and evaluation of HistGradientBoosting, LightGBM, and CatBoost classifiers. Task 4 (Temporal Data Mining): Apriori association rule mining on discretised temporal states, K-Means temporal clustering on cyclical features, and weekly time-series decomposition (additive and multiplicative).
3) **Ahilnandan Kabilan** – Task 5 (Joint Spatio-Temporal Inference): cross-feature engineering (ST_ZoneSeverity, RushHour×Intersection, Night×LowVis), joint Random Forest and LightGBM models, feature subset ablation study, five-fold cross-validation, ST-DBSCAN spatio-temporal clustering, and traffic-signal×night interaction analysis. Task 6 (Visualisations): all publication-ready figure panels and the interactive Folium heatmap.

Report written jointly by all three members.

Declaration of GenAI Usage

Claude (Anthropic) was used to assist with LaTeX formatting, report structure, expanding mathematical definitions and formal notation, and ensuring all assignment rubric questions were addressed. Google Gemini was used for proofreading and grammar checking of selected sections. GitHub Copilot was used for minor code completion during notebook development. All numerical results, model training and evaluation code, analytical interpretations, and research conclusions are the authors' own work.

GitHub Repository: Spatio-Temporal Data Analysis